

Junior Backend Engineer Assignment

Tech Stack

- Node.js
 - Express.js
 - MongoDB
 - TypeScript
-

Overview

Build a backend service called **TaskFlow API**.

The system should allow authenticated users to:

- Register/Login
 - Create tasks
 - Update tasks
 - Fetch tasks
 - Filter and search tasks
-

Functional Requirements

1. Authentication Module

Register User

Endpoint

`POST /api/auth/register`

Request Body

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "123456"
}
```

Requirements

- Email must be unique
 - Password must be hashed
 - Proper request validation
-

Login User

Endpoint

POST /api/auth/login

Request Body

```
{
  "email": "john@example.com",
  "password": "123456"
}
```

Response Example

```
{
  "success": true,
  "token": "jwt_token"
}
```

Requirements

- JWT-based authentication
 - Proper HTTP status codes
 - Invalid credentials handling
-

2. Task Module

Task Schema

```
{
  title: string;
  description: string;
  status: "TODO" | "IN_PROGRESS" | "DONE";
  priority: "LOW" | "MEDIUM" | "HIGH";
  dueDate: Date;
  createdBy: ObjectId;
  createdAt: Date;
  updatedAt: Date;
}
```

Create Task

Endpoint

```
POST /api/tasks
```

Requirements

- Protected route
 - Validate request payload
 - Store task in MongoDB
-

Get All Tasks

Endpoint

```
GET /api/tasks
```

Features

- Pagination
- Filtering
- Sorting
- Search by title

Example Queries

```
/api/tasks?page=1&limit=10  
/api/tasks?status=DONE  
/api/tasks?priority=HIGH  
/api/tasks?search=bug  
/api/tasks?sortBy=createdAt&order=desc
```

Get Single Task

Endpoint

```
GET /api/tasks/:id
```

Requirements

- Return proper error if task does not exist
-

Update Task

Endpoint

```
PUT /api/tasks/:id
```

Requirements

- Only task creator can update the task
-

Delete Task

Endpoint

DELETE /api/tasks/:id

Requirements

- Only task creator can delete the task
-

3. Validation & Error Handling

The application should include:

- Centralized error handling
- Request validation
- Proper HTTP status codes
- Meaningful error messages

Example Error Response

```
{
  "success": false,
  "message": "Task not found"
}
```

4. Technical Expectations

Please ensure:

- Proper TypeScript usage
 - Clean and organized project structure
 - Environment-based configuration
 - Maintainable and readable code
-

5. Bonus Features (Optional)

You may implement any of the following:

- Swagger/OpenAPI documentation
 - Docker setup
 - Unit tests
 - Aggregation APIs
 - Logging
 - Rate limiting
-

Submission Requirements

Please submit:

- GitHub repository
 - README.md
 - Postman collection
 - `.env.example`
-

README Expectations

Your README should include:

- Setup instructions
 - Environment variables
 - API documentation
 - Assumptions made
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Expected Completion Time

Estimated completion time:

24 hours